

TUCSON, AZ, USA

WMO: 722740

Lat: 32.228N

Lon: 110.956W

Elev: 2452

StdP: 13.44

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	32.0	34.7	-1.2	5.6	62.2	3.1	7.0	63.2	24.7	59.4	21.8	59.0	4.8	140	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	105.7	65.9	103.6	65.7	101.5	65.4	72.6	87.7	71.8	87.2	71.2	86.9	10.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	118.0	76.7	68.1	113.3	77.1	66.9	108.4	77.8	37.8	87.2	37.1	86.9	36.5	86.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.7	16.5	26.4	109.8	3.5	2.5	23.9	111.7	21.8	113.2	19.8	114.6	17.2	116.4		
(5)			21.7	74.3	3.6	1.1	19.1	75.2	17.0	75.8	15.0	76.5	12.4	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.6	53.7	56.1	62.3	67.9	76.5	86.3	87.9	86.4	82.5	72.2	61.7	52.9		
	DBStd	14.05	6.28	6.61	6.77	6.93	6.04	4.91	4.34	3.97	4.23	6.65	7.18	6.42		
	HDD50	116	37	20	5	1	0	0	0	0	0	0	7	46		
	HDD65	1328	353	255	130	50	3	0	0	0	0	20	141	376		
	CDD50	7635	150	190	385	539	821	1089	1175	1128	976	688	359	135		
	CDD65	3373	1	5	45	138	359	639	710	663	526	243	43	1		
	CDH74	47252	54	179	860	2071	5192	10034	10118	8827	6492	2763	620	42		
	CDH80	27051	3	25	234	856	2882	6594	6244	5174	3606	1273	159	1		
(14) Wind	WSAvg	7.3	6.7	7.0	7.5	8.0	7.9	7.8	7.6	7.1	7.4	7.3	7.0	6.5		
(15) Precipitation	PrecAvg	11.7	1.0	0.9	0.8	0.4	0.2	0.2	1.9	2.2	1.3	0.9	0.7	1.1		
	PrecMax	19.7	5.6	3.8	2.4	1.6	0.8	2.0	7.6	5.1	4.4	5.8	1.8	7.3		
	PrecMin	6.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0		
	PrecStd	3.4	1.0	0.9	0.7	0.4	0.2	0.4	1.4	1.2	1.1	1.3	0.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	83.5	89.8	94.8	102.6	109.5	108.4	106.6	102.4	96.7	88.2	78.4		
		MCWB	51.9	52.7	55.9	57.5	62.6	65.5	66.4	68.1	66.1	62.7	56.8	52.1		
	2%	DB	75.2	79.2	85.4	91.3	99.0	105.9	105.5	103.5	99.7	93.6	84.4	74.5		
		MCWB	50.5	51.6	54.4	56.5	60.6	64.2	67.1	67.9	65.7	61.5	55.3	51.3		
	5%	DB	71.7	75.7	82.4	88.5	96.2	103.8	103.1	100.9	97.7	90.8	81.1	70.5		
		MCWB	49.0	50.5	53.3	55.4	59.3	63.7	67.4	68.3	65.3	60.3	54.7	49.8		
	10%	DB	67.9	71.4	79.2	85.3	93.5	101.7	100.4	98.2	95.2	87.7	77.4	66.7		
		MCWB	48.1	49.3	52.0	54.4	58.1	62.8	67.5	68.4	65.5	59.2	53.6	48.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	57.2	58.9	60.4	65.9	71.1	73.4	73.8	73.0	68.3	60.1	56.5		
		MCDB	66.0	69.3	80.2	88.7	93.9	91.6	87.6	88.5	87.8	80.7	72.7	66.2		
	2%	WB	53.7	55.0	56.6	58.3	63.1	69.2	72.4	72.8	71.4	65.8	58.3	54.0		
		MCDB	66.7	68.9	79.0	85.9	91.2	92.1	87.2	88.2	84.6	81.8	75.0	67.6		
	5%	WB	51.7	53.1	54.8	56.7	61.1	67.8	71.7	72.0	70.3	63.8	56.6	52.1		
		MCDB	65.6	68.0	76.9	83.8	90.0	91.7	86.8	87.9	84.3	82.4	75.4	65.9		
	10%	WB	50.0	51.2	53.1	55.1	59.5	66.2	71.1	71.3	69.2	62.1	54.9	50.1		
		MCDB	64.4	67.6	75.2	82.1	89.4	93.2	86.8	87.5	84.1	81.6	74.2	64.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.5	25.7	27.5	28.6	29.2	28.7	22.8	22.4	23.7	26.4	26.4	24.6		
		MCDBR	30.9	32.3	32.6	32.5	32.3	30.7	26.3	25.9	26.7	29.8	30.8	29.4		
		MCWBR	14.7	14.7	13.6	13.3	12.6	10.8	7.0	5.9	7.4	11.3	13.4	14.1		
	5% WB	MCDWR	24.2	25.7	28.7	29.6	29.5	25.2	21.3	21.4	20.7	23.0	25.6	24.4		
		MCWBR	12.3	11.8	11.8	12.6	11.6	8.4	4.9	5.2	5.8	8.8	11.7	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.267	0.289	0.299	0.311	0.333	0.392	0.379	0.338	0.300	0.273	0.262		
	taud		2.632	2.625	2.533	2.488	2.492	2.437	2.403	2.428	2.515	2.566	2.599	2.627		
	Ebn at Noon		311	319	316	314	309	300	282	285	295	301	304	305		
	Edh at Noon		24	27	31	34	35	36	37	36	32	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		1138	1436	1909	2347	2599	2609	2177	2056	1882	1602	1260	1041		
	RadStd		80	115	93	73	95	136	132	72	90	100	61	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-146	+286	+122
(47) Regional (13 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-135	+187	N/A

Nomenclature: See separate page